

Vaibhav Pandey

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Profile

Computer Science undergraduate (8.86 CGPA) with hands-on experience in Software Engineering, Full-Stack Development and Applied AI. Experienced in developing and deploying functional applications, including real-time maritime monitoring dashboards, drone navigation systems (ROS/C++) and machine learning pipelines. Practical background in system reliability, cloud hosting and automated testing, supported by competitive programming and leadership experience.

Education

BMS Institute of Technology & Management <i>B.E. in Computer Science & Engineering</i> CGPA: 8.86	Bengaluru, Karnataka 2023 – 2027
Kendriya Vidyalaya No.1, AFS Jalahalli West <i>Class XII (Computer Science)</i> 93.60%	Bengaluru, Karnataka 2022 – 2023
Kendriya Vidyalaya No.2, AFS Ambala Cantt <i>Class X (Advanced Mathematics)</i> 94.20%	Ambala, Haryana 2020 – 2021

Technical Skills

Programming & Tools: Python, C/C++, SQL, PostgreSQL, MongoDB, JavaScript, HTML/CSS, Git, Docker, Kubernetes, AWS, Linux
AI & Analytics: Machine Learning, Predictive Modeling, Anomaly Detection, Geospatial Intelligence, OpenCV, TensorFlow, RAG
Autonomous Systems: ROS, PX4, Gazebo, Sensor Fusion, Visual-Inertial Odometry (VIO), Path Planning, MATLAB

Leadership & Extracurriculars

Founder & President Sep 2023 – Present
ASTRA Club (BMSIT&M) Bengaluru, Karnataka

- Founded and lead the university's premier technical defense club, fostering student interest in strategic technologies.
- Spearheaded the organization of the **Defence Technology Expo 2026**, providing a platform for students to showcase innovations in robotics, AI and aerospace to industry experts.
- Organized technical workshops and bridged the gap between academia and defense industries like DRDO, HAL and BEL.

Projects

Argo FloatChat AI | *Python, PostgreSQL, Pandas, Scikit-learn, REST APIs* [🔗 GitHub](#)

- Engineered** an AI-driven analytical model specialized for oceanographic datasets (NETCDF) achieving a response time of **<1s**.
- Developed** dynamic visual representations to simplify the interpretation of complex oceanic data.
- Designed** a highly scalable data pipeline capable of processing **60GB+** of complex datasets while maintaining optimal query performance.

Maritime Situational Awareness Dashboard | *JavaScript, Node.js, Express, React.js, Cloud Hosting* [🔗 GitHub](#)

- Developed** a real-time vessel tracking and geolocation dashboard monitoring 20,000+ vessels across India's 200-NM maritime boundary.
- Implemented** an anomaly detection pipeline using Isolation Forest, achieving **98% precision** in flagging simulated trajectory deviations and spoofed AIS streams.
- Deployed** the application on Render with automated health monitoring, maintaining **100% monitored uptime** over 44+ days and an average response latency of **610ms**.

AURA-X Autonomous UAV | *ROS, Python, C++, OpenCV, Gazebo, PX4, TensorFlow* [🔗 GitHub](#)

- Designed** an autonomous reconnaissance UAV capable of GPS-denied navigation utilizing Visual-Inertial Odometry and sensor fusion (combating spoofing/jamming).
- Developed** intelligent obstacle avoidance and adaptive path-planning algorithms for complex flight operations.
- Integrated** real-time localization and control within a modular PX4 architecture, reducing hardware procurement costs by **80%** using consumer-grade sensors while maintaining stable state estimation.

Supply Chain AI Trust Model | *Python, Machine Learning, Scikit-learn* [🔗 GitHub](#)

- Built** a machine learning pipeline to evaluate the trustworthiness of autonomous AI agents in supply chain operations, outputting confidence metrics for automated decisions.
- Implemented** ensemble classification algorithms using Scikit-learn to analyze decision anomalies and flag high-risk transaction patterns.
- Designed** validation scripts to test model robustness under noisy operational inputs, achieving an accuracy of **92%** during testing.

Awards & Achievements

NDA AIR 31: Secured All India Rank 31 in National Defence Academy (NDA) 153rd Course (Fighter Pilot Branch, IAF).
TES AIR 50: Secured All India Rank 50 in Technical Entry Scheme (TES) 52nd Course (Indian Army).
National-Level Hackathons: Track Prize at ANVESHANA; 2nd Place at InCSEption Inter-College Hackathon.
Sports: National-Level Basketball Player and Regional Gold Medalist.